

Siddarth Velnati

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EDUCATION

Purdue University
B.S. Computer Engineering
Certificate in Entrepreneurship and Innovation (Minor)

West Lafayette, IN
May 2028
GPA: 3.30/4.00

PROFESSIONAL EXPERIENCE

ARC Purdue Indy **Indianapolis, IN**
Robot Software Dev/Lidar Tester **Oct 2024 – Present**

- Led testing and validation efforts for the autonomous racing platform and coordinated with 6 to 8 team members to diagnose sensor, networking, and system-level issues
- Assembled and maintained the hardware platform for the racecar, including sensor mounting, power distribution, cable routing, and VESC integration, and ensured reliable operation during test runs
- Collaborated across perception, controls, and hardware sub-teams to integrate software updates with real-world testing schedules and ensure reliable system performance

Boiler Robotics **West Lafayette, IN**
Communication and Intelligence/Robotic Arm Sub team Member **Jan 2025 – Present**

- Developed and tested software modules for a semi-autonomous rover on an NVIDIA Jetson platform, including sensor integration, motor control, and real-time data handling
- Organized field tests and sensor checks for the rover platform, verified hardware functionality, and helped maintain consistent testing procedures for the team

Qrati Labs **Boulder, CO**
AI Dev/Design Intern **Oct 2023 – Mar 2024**

- Collaborated with a cross-functional engineering team to integrate generative AI features into a production content platform, supporting data workflows for more than 100 users
- Streamlined prompt-engineering and evaluation processes and helped develop reproducible model-usage guidelines, improving response quality and documentation clarity

Kumon North America Inc. **Livermore, CA**
Grader/Senior Tutor **Sep 2025 – Feb 2023**

- Mentored and supported more than 40 students, helping them strengthen analytical and problem-solving skills through individualized instruction
- Reviewed and graded over 100 assignments per shift with accuracy and maintained organized progress records to support instructor evaluations

PROJECTS

F1TENTH Autonomous Racing Vehicle | ARC Purdue Indy

- Implemented LiDAR preprocessing and filtering pipelines and integrated ROS2 drivers on a Jetson platform to enable real-time obstacle detection at 20–30 Hz
- Developed mapping, localization, and control modules (PID and pure pursuit) and tested them through large simulation batches in F1TENTH Gym
- Integrated LiDAR, IMU, encoders, and the VESC into a unified ROS2 system and created reliable hardware bring-up procedures for full-stack testing

Social Media Market Sentiment Analyzer

- Designed a data pipeline using PRAW and NLTK to automatically scrape, clean, and preprocess over 100 Reddit posts for sentiment analysis
- Applied VADER sentiment scoring to extract polarity trends and identify high-impact market discussion patterns
- Built an interactive Steamlit dashboard with Plotly visualizations to display sentiment trends, keyword frequency, and real-time market insights

Random Walker Simulation Study | University of California, Davis

- Developed Brownian motion and random walker simulations in Python and C to model molecular behavior inside living cells, running more than 500 trials to analyze particle movement patterns
- Applied probability theory and computational methods to study how simple stochastic interactions can lead to organized molecular behavior in biological systems

SKILLS

Programming/Perception: Python | C/C++ | ROS2 | Bash | LiDAR Processing | Sensor integration | SLAM | PID & Pursuit Control | PyTorch

Hardware: NVIDIA Jetson | Embedded Linux | Serial Communication | Power Distribution | Basic Electronics | Soldering

Leadership & Service: Eagle Scout, Boy Scouts of America | Project Leadership | Peer Mentorship | Technical Presentations